

Exercise 1: A simple greeting script

I opened my text editor

```
(root@kali)-[~]
# l3afpad
```

And inputted the following;

```
greetingscript
File Edit Search Options Help
#!/bin/bash
# Greeting Script

echo "what is your name?"
read name

echo "Hello, $name! Welcome to bash scripting,Keep hacking and learning!"
```

After exiting the text editor, In the terminal, I did a “ls -l” command and found out that the file I most recently created “**greetingscript**” does not have the execution permission.

```
(root@kali)-[~]
# ls -l
total 48
-rwxr-xr-x 1 root root 317 Feb 16 00:03 advancedportscanner
-rw-r--r-- 1 root root 186 Feb 15 23:35 'a simplified scanner script'
-rw-r--r-- 1 root root  0 Feb  1 07:21 bede
-rw-r--r-- 1 root root 142 Feb 17 00:45 greetingscript
```

In order to change this, I did a “chmod 755” command

```
(root@kali)-[~]
# chmod 755 greetingscript
```

Which gave me permission to execute the file.

```
(root@kali)-[~]
# ./greetingscript
what is your name?
chukwuka
Hello, chukwuka! Welcome to bash scripting,Keep hacking and learning!
```

Exercise 2: script a scanner

Here I scripted a scanner to find systems with Microsoft's SQL database at port 1433.

```
MSSQL
File Edit Search Options Help
#!/bin/bash
# Microsoft SQL Scanner

echo "Enter the starting IP address?"
read FirstIP

echo "Enter the last octet of IP address?"
read lastoctetIP

echo "enter port number you want to scan for?"
read port

echo "Scanning $IP on port $port ..."
nmap -sT $FirstIP-$lastoctetIP -p $port >/dev/null -oG MSSQL1

cat MSSQL1 | grep open > MSSQL2
cat MSSQL2
```

From the screenshot above, I used the **echo** command to prompt the user to input the value the variables would hold. Next, I used the **read** command to capture the input from the user. Next, I sent the standard output from the **nmap** command to **/dev/null**, then I sent the output in a **grep**-able format to a file named **MSSQL1**. Next, I used the **cat** command to read the file contents of **MSSQL1** and used the **grep** command to search for the word “open” and then transfer the output into a new file named **MSSQL2**.

I then used the **ls -l** command to display permissions on recent files and discovered my **MSSQL** file did not have the execute permission.

```
rw-r--r-- 1 root root 88 Feb 13 13:37 netconack
rw-r--r-- 1 root root 0 Feb 1 07:21 life
rw-r--r-- 1 root root 0 Feb 1 07:21 love
rw-r--r-- 1 root root 356 Feb 17 06:59 MSSQL
```

In order to have the execute permission, I used the **chmod 755** command

```
(root@kali)-[~]
# chmod 755 MSSQL
```

Here, I successfully executed the file

```
(root@kali)-[~]
# chmod 755 MSSQL

(root@kali)-[~]
# ./MSSQL
Enter the starting IP address?
192.168.56.1
Enter the last octet of IP address?
1
enter port number you want to scan for?
1433
Scanning on port 1433 ...
```

I checked out the newly created **MSSQL1** & **MSSQL2** file from the execution

```
*MSSQL1
File Edit Search Options Help
# Nmap 7.95 scan initiated Tue Feb 17 07:38:05 2026 as: /usr/lib/nmap/nmap -sT -p 1433 -oG MSSQL1 192.168.56.1-1
# Nmap done at Tue Feb 17 07:38:08 2026 -- 1 IP address (0 hosts up) scanned in 3.16 seconds
```

```
*MSSQL2
File Edit Search Options Help
```

Conclusion

From the screenshot above the port **1433** is not open(closed).